A nuclear-register criterion for radical-pair computation in cryptochrome

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Cryptochrome, the protein behind the magnetic compass of migratory birds, hosts a pair of radicals whose electron spins stay coherent long enough to sense the geomagnetic field. It has long been conjectured that the same spin chemistry could support computation in a nervous system, but the conjecture has never been stated in a form that chemistry can test. Here we state and evaluate one. We ask whether the spin system of a cryptochrome-type radical pair can carry information from one chemical cycle to the next, and whether ordinary chemistry can read it back. Solving the pair's spin dynamics with recombination and decoherence, we find that it can: the memory is stored in the nuclear spins, which survive recombination in the product, so the relevant clock is the interval between chemical cycles rather than the microsecond lifetime of the pair, and the product's nuclear polarisation reads the memory out. This yields a criterion with three measurable constants: the interval between cycles, the relaxation time of the product's protons at the geomagnetic field, and the number of molecules read as one pool. Predicting the relaxation time shows that a protein-bound proton forgets in milliseconds, which caps the memory at about one to three milliseconds, below the timescale of neural activity unless the register reorients several times faster than the protein. No quantum advantage over a matched classical network survives a fair comparison. All three constants can be measured in vitro.

I. INTRODUCTION

A magnetic field as weak as the Earth's, about 50 μT, can change the outcome of a chemical reaction through the radical-pair mechanism[1,2,16]: two radicals born with correlated electron spins interconvert coherently between singlet and triplet under the hyperfine fields of nearby nuclei, and because recombination depends on the spin state, a field that retunes the interconversion changes the product yields. The mechanism underlies the leading model of avian magnetoreception, in which the photoreceptor cryptochrome hosts a flavin–tryptophan radical pair[2–5,12]; only a spatially separated pair, with a microsecond lifetime and weak electron–electron coupling, keeps the coherence that weak-field sensitivity requires[23].

The success of the mechanism has repeatedly prompted a bolder conjecture: that spin coherence in biomolecules might support not merely sensing but information processing, a "quantum brain"[6,7], contested on decoherence grounds[9] and part of the broader programme of quantum biology[10,11]. The conjecture has never been stated in a form that chemistry can test: no proposal names the quantity a downstream reaction would read, the clock on which it is read, or the relaxation constant that bounds it.

This paper states and evaluates such a form. We ask one question: *can the spin system of a cryptochrome-type radical pair carry information from one chemical cycle to the next, and can ordinary chemistry read it back?* We answer it by simulating the pair's spin dynamics, with recombination and decoherence, through many successive cycles driven by different inputs, and asking how much of the past input sequence the chemical products of the current cycle still encode. The framework is reservoir computing[13,24,25]: a fixed dynamical system is driven by a time series and a linear readout of its state is trained to reconstruct past inputs. Its central quantity, the memory capacity, is the number of such functions the readout recovers; its central requirement, fading memory, is supplied here by the openness of the chemical cycle, which discards the electron pair every turnover and keeps only the nuclei. Table I translates each term into the chemistry it stands for, and Fig. 1 shows the cycle.

Three results follow. First, the readout is not the obstacle: the time course of a product and, above all, the nuclear polarisation the product carries away (chemically induced dynamic nuclear polarisation, CIDNP[18]) recover memory well above their floors, with no spectroscopy. Second, the memory is nuclear and survives recombination, so the clock is the turnover interval, not the microsecond pair lifetime of the usual timescale objection; what replaces that objection is the *register-reuse criterion*: the same nuclear register must be reused from one turnover to the next and must survive the interval. Third, predicting that register's relaxation at the geomagnetic field, a protein-bound proton forgets in milliseconds, which caps the memory horizon below the band of neural rhythms unless the register reorients four to fifteen times faster than the protein. No quantum advantage over a matched classical network survives a fair comparison, and the drive cannot be light.

We state at the outset the load-bearing caveat. Whether a *neuronal* cryptochrome forms a separated radical pair at all is unestablished: the mammalian CRY1/CRY2 proteins are primarily circadian transcriptional regulators[15]. Every biological conclusion below is conditional on the existence of such a pair; the spin-dynamics results stand on their own. The question arose from a layered proposal[19,20] of a nuclear-spin memory

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TABLE I. The correspondence used throughout: each term of reservoir computing, what it is in the radical pair, and how it is measured here.

term	in the radical pair	measured as
input	the singlet character of the pair at birth (alternatives: field magnitude, recombination rate)	one value per chemical cycle
reservoir state	the nuclear spins (the *register*), which outlive the electron pair	nuclear density matrix handed from one cycle to the next
readout	quantities a downstream reaction could use: product yields over time; the nuclear polarisation the product carries away	linear fit from these to past inputs
memory capacity	how many past inputs, and functions of them, the readout reconstructs	sum of squared out-of-sample correlations
memoryless floor	what the same readout scores with the register wiped every cycle	same estimator, register randomised
excess	capacity above the floor: the memory itself	difference of the two
clock	the interval between successive chemical cycles	T_{turnover}, not the pair lifetime τ
memory horizon	capacity above the zero-delay term, times the clock	$(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$

read through a radical-pair interface; this paper is its quantitative test, and what is new relative to it is the chemically consistent drive, the restriction of the readout to what chemistry can carry, the matched classical baselines, and the turnover clock with the criterion that follows. The spin Hamiltonian, the Haberkorn treatment and the capacity estimators are standard[17,30–32].

Section II describes the model and the measures in the terms of Table I. Section III presents the results in the order above, Sec. IV states what would test the criterion and where the model is limited, and the appendix gives the equations.

II. THE MODEL IN BRIEF

The spin system. The pair is modelled as five spin-1/2 particles, the two electrons and three nuclei (two ^{14}N on the flavin, one proton on the tryptophan) with

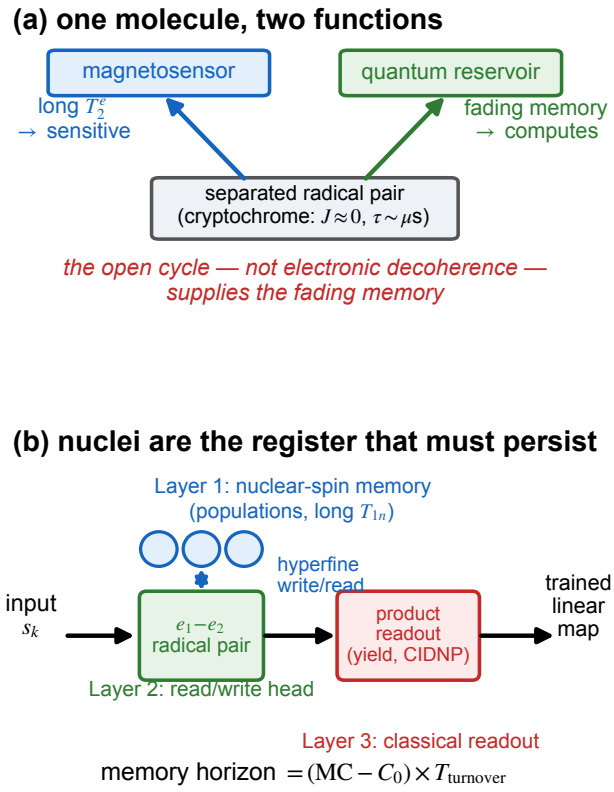


FIG. 1. **The chemical cycle as a reservoir.** (a) A cryptochrome-type separated radical pair is one molecule in two roles: its long electronic coherence makes it a weak-field magnetosensor, and the dissipation of an open cycle supplies the fading memory a reservoir needs. (b) The nuclei are the register. They are not consumed by recombination and persist from one turnover to the next, whereas the electron pair is created afresh each cycle, so the memory horizon $(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$ is set by the turnover interval and not by the pair lifetime τ. It reaches the neural band only if the register is reused *and* survives the pause, which requires it to be carried by protons and to reorient with $\tau_c \lesssim 2.6$–4.0 ns (supplementary material, Secs. S9 and S12).

the couplings of cryptochrome's flavin–tryptophan pair (supplementary material, Sec. S13), evolving under the Haberkorn master equation[17,30] with spin-selective recombination and electronic dephasing (Appendix A). The *cryptochrome point* ($B = 50$ μT, lifetime $\tau = 1$ μs, no electron–electron coupling as the reference premise) is where every chemical readout, the clock and the criterion are evaluated; an *engineered point* ($B = 1$ mT, $\tau = 50$ ns, stronger couplings) is a molecular-spin ref-

erence used only for the full-readout capacity and the classical comparison.

The cycle. Each cycle the nuclei are retained and a *new* spin-correlated electron pair is born whose singlet character $s \in [0,1]$ encodes the input, $\rho_e(s) = s\,|S\rangle\langle S| + (1-s)P_T/3$; the pair evolves for its lifetime, recombines spin-selectively, and hands its nuclear register to the next cycle. This is how a turnover drives the system; the single-electron reset of earlier treatments is not a chemical process and, for a separated pair, leaves no capacity at all (supplementary material, Sec. S3). The register handed on is either that of the molecules not yet recombined at τ (survivor) or the one carried away by those that have (product, Eq. (A4)); the product register is the faithful one and gives larger capacities, so the survivor numbers reported are conservative.

The measures. The linear memory capacity MC[31] counts how many delayed inputs a linear readout reconstructs; the information processing capacity IPC[32] adds quadratic functions of them; both are out of sample. A readout with no memory still reconstructs the present input, so every route has a memoryless floor, measured by wiping the register every cycle, and only the excess above it is memory. Capacities are means over several input realisations (twelve for the clock scan; Appendix A).

III. RESULTS

A. Ordinary chemistry can read the reservoir

Biology has no spectrometer; it has products. We therefore restrict the readout to observables a downstream reaction could plausibly use, with recombination switched on so that the quantity read is the yield actually produced, $Y_X = k_X \int_0^\tau \mathrm{Tr}[P_X \rho(t)]\,dt$. Table II and Fig. 2 list all seven routes the model defines, each against its own floor.

The floors differ because IPC counts linear and quadratic functions of the present input as well as of past ones: a readout with several channels reconstructs both s and s^2 of the current cycle with no memory at all, so its floor is about 2, whereas a yield route's floor is 1. The route labelled three micro-environments reads three copies of the molecule with different lifetimes in parallel. Three things follow. A single end-point yield remembers nothing: its capacity equals its floor. The time course of one product does remember, clearing its floor by 0.95, and adding the second exit channel buys only 0.003 more, so the number of product observables is not what limits the system. And the nuclear polarisation the product carries away is the best readout by a wide margin, 4.42 against a floor of 2.00, with no spectroscopic access. The floors differ between routes, so a universal floor of 1 would overstate the richer readouts, and the accumulated-pool route is excluded from every claim because a leaky integrator is itself a delay line. The mechanism is spin sorting: either electron of a newly born pair is maximally mixed for every input, so the input lives in the two-electron correlation, and only spin-selective recombination converts it into a nuclear population difference (with recombination off the capacity is the floor, 1.02); with the electron–electron coupling the source reports[23] the interacting electrons write too (Sec. III F).

With the product register (Eq. (A4)) the same readouts are stronger: time-resolved kinetics rise from 1.97 to 2.77 and the CIDNP route from 4.42 to 4.96 (Table III), so the survivor numbers reported above are the conservative ones. Every result below that *reduces* the claims has been re-run with the product register and survives (supplementary material, Sec. S4).

B. The memory is nuclear, so the clock is the turnover

The horizon over which a reservoir remembers is its capacity multiplied by its cycle time. Identifying that cycle time with the pair lifetime $\tau \sim 1$ μs would put the system 10^3–10^5 below neural timescales, but the memory is carried by the nuclei, which leave in the diamagnetic product and relax slowly; the cycle time is the interval between turnovers.

Two definitions are needed. The horizon is quoted as $(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$, since the zero-delay term $C_0 \simeq 1$ reports the present input and would otherwise assign a growing horizon to a system with no memory. And it is an integrated delay capacity, not the longest delay reconstructed; for the CIDNP readout the capacity by delay is 1.00, 0.85, 0.47, below 5% of C_0 by four turnovers, so $\mathrm{MC} - C_0 = 1.82$ of delayed capacity in that six-seed run (the twelve-seed clock scan of Table IV gives 1.92), and no quoted horizon hides a long tail (supplementary material, Sec. S2). Unless stated otherwise, capacities in this and the next section are linear memory capacities MC of the eight-channel CIDNP route with the survivor register, whose floor is 1.00; the IPC values of Table II include the quadratic terms and have the floor 2.00.

Inserting a pause T_d between turnovers and relaxing the register with probability $q = 1 - \exp(-T_\mathrm{d}/T_{1n})$, with $T_{1n} = 1$ s as a placeholder (the value measured at high field; Sec. III D replaces it by the geomagnetic-field value, which is what makes the horizon fall short), gives Table IV and Fig. 3(a).

With the same seeds at every T_d, the change is assessed on paired differences: across five decades the capacity falls by 0.239 ± 0.028 (SEM, $n = 12$, $t = 8.46$), that is 8.2%, resolved well above the seed noise and the time-grid error (0.03% from $n_t = 96$ to 192) and concentrated in the last decade. A single-realisation scan cannot resolve a change of this size; it is retained as supplementary material, Table S11, whose single realisation is why it is a control and not a result. A 10^5-fold change in the clock moves the capacity by under a tenth: the horizon tracks the turnover interval, and a turnover of 10–100 ms places it at 19–168 ms, inside the neural band, *provided the register survives the pause.* That proviso is what the rest of

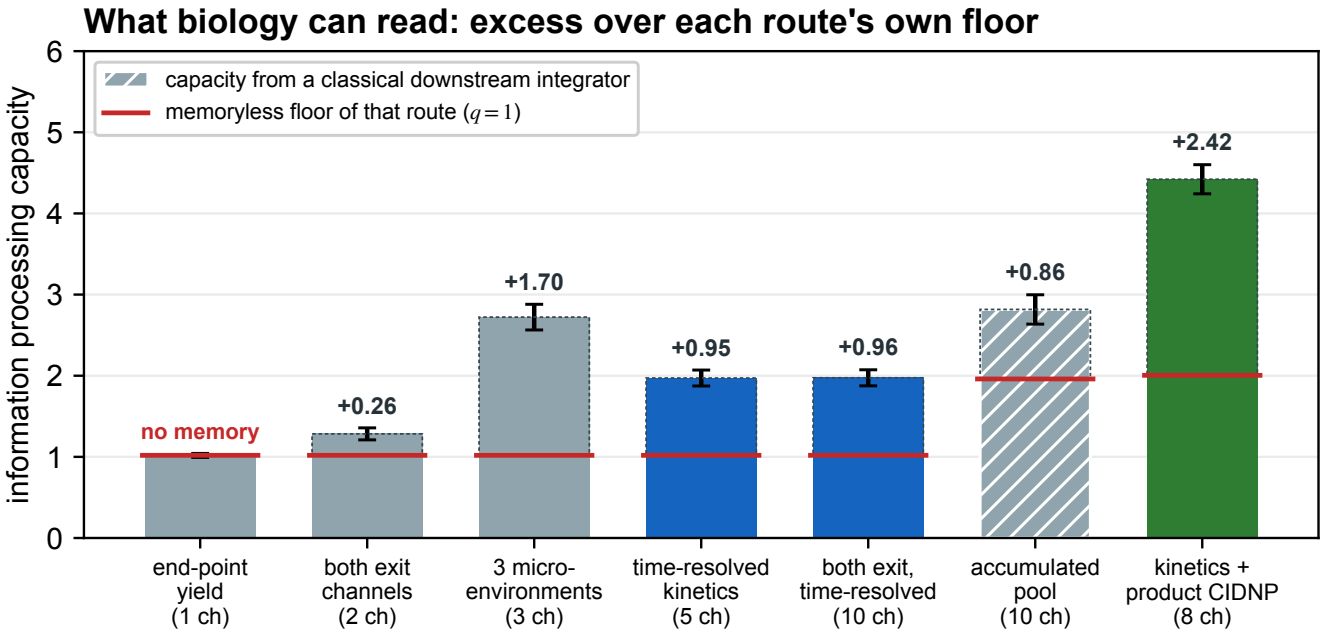


FIG. 2. **Readout is not the bottleneck.** Capacity of the chemically accessible routes at the cryptochrome point ($B = 50\ \mu$T, $J = 0$, $\tau = 1\ \mu$s). Red bars are each route's own memoryless floor; the excess printed above each bar is what measures memory. Kinetics clear their floor by 0.95, the product's nuclear polarisation (CIDNP) by 2.42, the end-point yield by nothing. The accumulated pool is hatched because a leaky integrator is itself a delay line.

TABLE II. Capacity recovered by chemically accessible observables at the cryptochrome point, each scored against its own memoryless floor.

readout route	ch.	IPC	floor	excess
end-point singlet yield	1	1.02	1.02	0.00
both exit channels, end point	2	1.28	1.02	0.26
three micro-environments (different τ)	3	2.72	1.02	1.70
time-resolved kinetics, one product pool	5	1.97	1.02	0.95
both exit channels, time-resolved	10	1.97	1.02	0.96
kinetics + CIDNP on the product	8	**4.42**	2.00	**2.42**
kinetics + accumulated downstream pool	10	2.82	1.96	0.86

TABLE III. Capacity with the survivor register against the physically faithful product register.

readout route	ch.	survivor	product
end-point singlet yield	1	1.02	1.01
time-resolved kinetics	5	1.97	**2.77**
both exit channels, time-resolved	10	1.97	2.78
kinetics + CIDNP on the product	8	4.42	**4.96**

the paper is about.

C. The criterion: the same register must be reused

Everything above assumes that the register written into is the *same* register on the next turnover. Depolarising the nuclei with probability q between cycles gives Fig. 3(b) (four seeds; the twelve-seed value at $q \to 0$ is 2.92): the eight-channel MC falls from 2.85 at $q = 0$ to 2.33 at $q = 0.3$, 2.09 at $q = 0.5$, 2.00 at 0.7 and 1.93 at 0.9, and at $q = 1$, a fresh molecule every cycle, to the floor MC = 1.00.

Which relaxation process enforces this can be asked directly. Replacing the depolarising channel by pure dephasing, which destroys nuclear coherence but preserves $\langle I_z \rangle$, leaves the capacity within five per cent (Table V): the memory is carried by nuclear populations, exactly what CIDNP reports, and the relevant constant is T_{1n}.

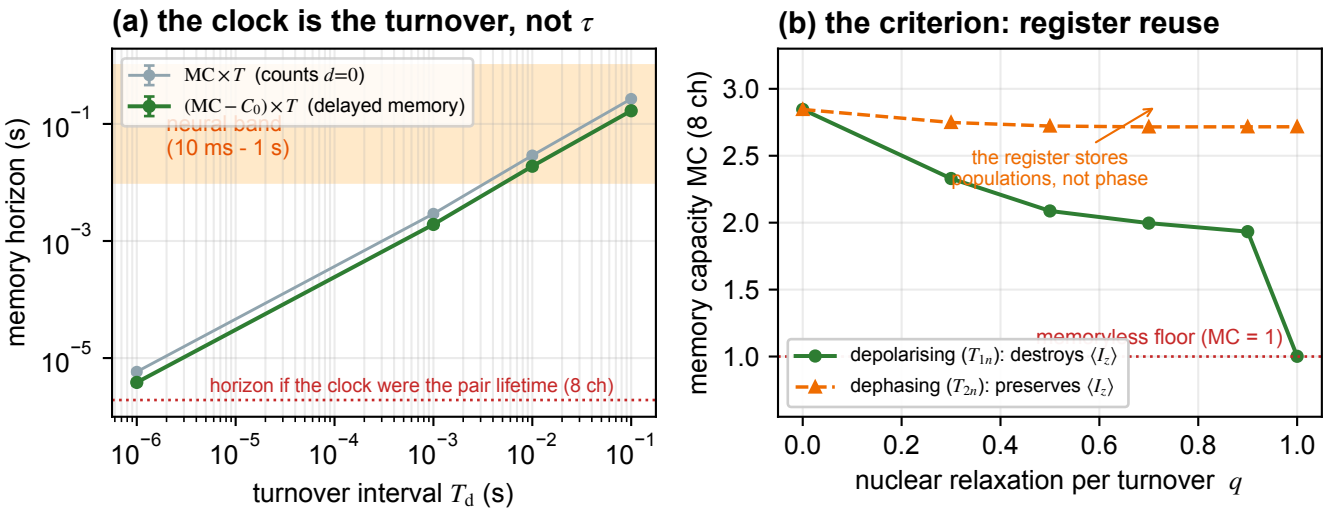


FIG. 3. **(a)** The memory horizon against the turnover interval (twelve realisations; error bars, between-seed s.d.). A physiological pause costs only 8% of the capacity across five decades, so the horizon tracks the clock and enters the neural band (shaded, 10 ms–1 s, the range of cortical rhythms[8]). Grey, raw $\mathrm{MC} \times T_{\mathrm{turnover}}$; green, the delayed-memory horizon quoted in the text; dotted, the horizon if the clock were the pair lifetime. **(b)** Depolarising the register between turnovers with probability q (green) drives the capacity to the floor at $q = 1$; pure dephasing at the same q (orange) costs under five per cent, so the register stores populations and the bounding constant is T_{1n}.

TABLE IV. The memory horizon against the turnover interval, twelve input realisations.

T_{d}	q	MC (8 ch)	$\mathrm{MC} \times T$	$(\mathrm{MC} - C_0) \times T$
1 μs	10^{-6}	2.919 ± 0.205	5.8 μs	3.8 μs
1 ms	10^{-3}	2.916 ± 0.204	2.9 ms	1.9 ms
10 ms	10^{-2}	2.894 ± 0.196	29 ms	19 ms
100 ms	9.5×10^{-2}	2.679 ± 0.124	268 ms	168 ms

At the geomagnetic field, however, extreme narrowing ($\omega\tau_c \approx 2 \times 10^{-4}$) makes T_{1n} and T_{2n} coincide to one part in 10^7 (supplementary material, Sec. S12), so the distinction buys nothing at the operating point.

TABLE V. Which nuclear relaxation process enforces the criterion.

q	MC (8 ch), depolarising	MC (8 ch), dephasing
0.0	2.846	2.846
0.5	2.088	2.722
0.9	1.933	2.716
1.0	1.002	2.716

Not every nucleus can hold the register. Protons in proteins relax on the second timescale at NMR fields[34], but ^{14}N is quadrupolar (spin 1) and relaxes in $\sim$ 0.2–0.3 μs at the geomagnetic field (supplementary material, Sec. S12), so the two flavin ^{14}N are wiped every cycle. Restricting the register to the tryptophan proton leaves IPC = 2.78 against the IPC floor of 2.00, an excess of 0.78 where the full register gave 2.42, and the kinetic readout falls to its floor (supplementary material, Sec. S9): the criterion survives on a proton register, at 32% of the excess (29% with the product register) and through CIDNP alone. Scanning that register as the biology would keep it, ^{14}N wiped, product carried and the proton alone depolarised with probability q_{H}, gives an MC of 1.85 at $q_{\mathrm{H}} = 0$ that holds to 1.89 at 0.5 and falls through 1.67 at 0.8 and 1.30 at 0.9 to the floor 1.00 at $q_{\mathrm{H}} = 1$. This is the curve the criterion's numbers are read from.

The criterion is then the following. *The same nuclear register must be re-used across successive turnovers, and successive turnovers must be driven by distinct input samples. The register must be carried by protons, since the flavin ^{14}N cannot hold one. Given that, the memory horizon is* $(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$, *bounded above by the register's own T_{1n}; without it, the system retains nothing.* It is falsifiable by chemistry: it fails if the radical pair is regenerated on a fresh molecule each turnover, or if T_{1n} in the diamagnetic product is shorter than the turnover interval.

D. The criterion's constants, predicted

Neither T_{turnover} nor T_{1n} has been measured for a neuronal cryptochrome, but T_{1n} can be predicted from measured constants (supplementary material, Sec. S12), and the result is adverse. At the geomagnetic field nuclear relaxation is in the extreme-narrowing regime, where $1/T_1 \propto \tau_c$ is maximal, so the second-long T_{1n} used above is a high-field value: a proton on a 60 kDa protein relaxes in 2.4 ms at 50 μT against 9.0 s at 9.4 T. What sets the ceiling is the effective correlation time of the register's dipolar vector, $\tau_{\mathrm{eff}} = S^2\tau_c + (1 - S^2)\tau_{\mathrm{int}}$, with the dipolar sum over intramolecular and intermolecular partners.

Why the field matters deserves a sentence, because the result rests on it. A nuclear spin relaxes through the fluctuating magnetic field of neighbouring nuclei, sampled at its own Larmor frequency; the fluctuation is produced by molecular tumbling with correlation time τ_c (about 15 ns for a 60 kDa protein) and by faster local motion with correlation time τ_{int}, of which a fraction S^2 of the interaction is *not* averaged away. At 50 μT the proton Larmor frequency is 2 kHz, far below $1/\tau_c$, so the spectral density sits at its zero-frequency maximum and T_1 is as short as tumbling can make it; at 9.4 T it is sampled at 400 MHz, far above $1/\tau_c$, and T_1 is seconds. Protons of the surrounding water and protein (the intermolecular bath, calibrated here at 54% of the intramolecular rate on the water row of the validation table of supplementary material Sec. S12) add to the rate.

Feeding $q = 1 - \exp(-T_{\mathrm{d}}/T_1)$ with the predicted T_1 into the measured MC(q) curve and maximising the horizon over T_{d} gives the ceiling on the horizon, and solving for the τ_{eff} at which the ceiling reaches 10 ms gives the boundary (Table VI). Three registers are compared: the three-nucleus survivor register of Fig. 3(b), which assumes every nucleus can be kept; the proton-only product register the biology can keep; and that register with the molecule budget closed, either by regenerating only the third of the molecules that recombined by τ and replacing the rest with fresh ones, or by letting both branches continue, in which case their opposite polarisations largely cancel and the band is not reached at any τ_c. On the register the biology can keep the cap is 2.6 ms (1.7 ms with the bath), below the 10 ms edge of the band, and the band opens only if the register reorients 3.8 (5.9) times faster than the protein; closing the budget makes it 10 (15) times. The flavin 8α-methyl protons, whose intramethyl vector has $S^2 = 1/4$ and so is partly averaged by methyl rotation, relax more slowly ($T_1 = 5.0$ ms) and would clear the band on the full register, but not on the proton-only one. At a given τ_c the viable turnover intervals form a window (10–14 ms at 5 ns, 7.3–79 ms at 1 ns, 5.6–790 ms at 0.1 ns, bath included), so the two conditions are not independent.

The drive cannot be light: with the measured flavin extinction coefficient a single flavin turns over once per 7.7 s in full sunlight and once per some hundreds of years inside a skull, so a photochemical cycle fails by about ten orders of magnitude and the regeneration must be catalytic. The Hamiltonian is that of the photo-generated [$\mathrm{FAD}^{\bullet-}$ $\mathrm{TrpH}^{\bullet+}$] pair, and no dark, catalytic route that regenerates that pair on a cryptochrome is established; the criterion therefore names a requirement on whatever flavoenzyme cycle is proposed to meet it, not a cycle known to exist.

TABLE VI. The criterion's numbers at the geomagnetic field: the ceiling on the memory horizon for a register that reorients with the intact protein, and the τ_{eff} below which the horizon reaches the 10 ms band edge, without and with the intermolecular bath.

register	ceiling (ms)		τ_{eff} bound (ns)	
	dry	bath	dry	bath
3 nuclei, Trp Hβ	5.4	3.5	8.3	5.4
3 nuclei, methyl	11.2	—	17.1	9.6
proton, product, Trp Hβ	2.6	1.7	4.0	2.6
proton, product, methyl	5.4	3.0	8.2	4.6
proton, recombined third only	1.0	0.7	1.6	1.0
proton, both branches	none	none	none	none

E. There is no quantum advantage

A capacity is only meaningful against a classical reference read out symmetrically. We drive tanh echo-state networks (ESNs) with the same input and score them with the same estimator. At the engineered point, with a twelve-observable spin readout, the reservoir attains IPC $= 5.6 \pm 0.2$; matched by physical unit it is modestly ahead of a five-node network, matched by readout dimension well behind (Table VII), and which accounting is fair depends on a cost model we do not have.

TABLE VII. The quantum reservoir against matched classical reservoirs at the engineered operating point.

reservoir (readout features)	IPC
quantum, 5 spins (12 observables)	5.59 ± 0.19
ESN, 5 nodes (5 linear features)	4.55 ± 0.32
ESN, 5 nodes (5 linear + 7 quadratic)	9.41 ± 0.86
ESN, 12 nodes (12 linear features)	10.37 ± 0.59

At the cryptochrome point the comparison is channel-for-channel against the chemical readouts (Table VIII). With the ridge fixed at $\lambda = 10^{-6}$ on raw features the network wins by 57% at eight channels and 127% at five, but that margin belongs to the regulariser: the chemical channels have amplitudes of order 10^{-2} and the network states of order one, so one absolute ridge penalises the quantum readout far more. Standardising the features and choosing λ by cross-validation for both systems removes it: the paired difference is $+0.06 \pm 0.16$ (SEM,

$n = 12$) at eight channels and -0.10 ± 0.10 at five, that is $+1\%$ and -2%, neither resolved from zero. On no accounting does the radical pair compute better than a small classical network, and on no properly regularised accounting does it compute worse. A correlated-spin reservoir with full quantum readout has been reported to outperform classical reservoirs[46]; there is no tension, because what is read out differs. The advantage, where it exists, lives in the readout access, which is exactly what a chemical cycle does not grant.

TABLE VIII. The same comparison at the cryptochrome point, channel-for-channel. Left: ridge fixed at $\lambda = 10^{-6}$ on raw features (six seeds). Right: features standardised and λ chosen by five-fold cross-validation on the training half, for both systems (twelve paired seeds).

system (readout features)	λ fixed	λ by CV
quantum, kinetics (5 ch)	1.97 ± 0.10	4.64 ± 0.29
ESN, 5 nodes (5 features)	4.47 ± 0.33	4.55 ± 0.30
quantum, +CIDNP (8 ch)	4.42 ± 0.18	6.90 ± 0.47
ESN, 8 nodes (8 features)	6.94 ± 0.56	6.95 ± 0.61
ESN, 12 nodes (12 features)	9.62 ± 0.93	—

How much of the capacity is quantum at all is addressed in the supplementary material (Secs. S5 and S8) and is not needed for the criterion.

F. What the numbers do and do not depend on

Five sensitivities bear on the criterion; each is computed in full in the supplementary material.

The coupling premise. The cryptochrome point idealises "separated" as no electron–electron coupling, whereas the source reports $D = -11.4$ MHz and an exchange gap of 1.3–126 MHz of undetermined sign[23]. With those values (supplementary material, Sec. S14) the no-recombination capacity rises from the floor to 5.1–5.9 and the chemically accessible routes move as Table S31 records, with every floor unchanged; the ceiling rises from 3.4 to 5.7 ms with the bath, still below the band. The criterion survives the premise; its margin is what the premise sets.

The hyperfine tensors. With the published tensor set of the pair[12,21], orientation-averaged and with a fourth nucleus added (supplementary material, Sec. S11), the full register's CIDNP excess falls from 2.96 to 0.93, because near-axial nitrogens are written far less than isotropic ones, while the proton-only register the criterion rests on retains 0.60 against 0.86 (0.98–1.47 with a second proton). The headline capacities overstate the full register; the criterion's numbers are not loosened.

Regularisation and counting noise. The ridge $\lambda = 10^{-6}$ on unstandardised features is load-bearing for the kinetic route, whose capacity runs from 4.50 at $\lambda = 10^{-12}$ to 1.93 at 10^{-6} to 1.00 at 10^{-1}; the reported values are conservative with respect to it but not independent of it. The physically meaningful noise is that of counting a pool of N molecules, in which a product polarisation of order 10^{-2} is the mean of Ny_S values of ± 1 and so carries a counting variance of about $1/Ny_S$ that does not shrink with the polarisation. Counted that way (supplementary material, Sec. S7), the CIDNP excess is 2.41 of its noiseless 2.42 at $N = 10^{10}$, 1.93 at 10^8 and 0.49 at 10^6: the readout is counting-noise limited below about 10^9 molecules per pool, and the criterion carries a third constant, the number of molecules read as one pool.

Pooling and synchrony. Pooling $\sim 10^{10}$ copies is safe against heterogeneity of couplings and relaxation rates but not against desynchronisation, which destroys the kinetic readout's memory within a cycle while CIDNP retains 40–49% of its excess even at two cycles of jitter (supplementary material, Sec. S10); CIDNP is the surviving route under both the nuclide and the synchrony restriction.

What carries the input. No chemistry is named that modulates s over $[0, 1]$. Driving the reservoir through the field magnitude over 25–75 μT instead gives a CIDNP-route excess of 3.14 against 2.96, and through the recombination rate 2.42 (supplementary material, Sec. S3), so the criterion does not rest on a chemically unmotivated input; at a hundredth of the input range the excess falls to 0.76. Whether a neuron modulates the field is a separate question.

IV. DISCUSSION

What the criterion says. Three natural claims fail, though not as expected: a chemical readout is not too impoverished, because it need not be scalar; a few-spin reservoir does not beat a classical system of the same size at any accounting; and the microsecond lifetime does not put biological computation out of reach, because it is the wrong clock. The timescale gap moves rather than vanishes, to the nuclear relaxation time, which at the geomagnetic field is milliseconds. The criterion still compares two times, as the argument it replaces did; what the spin dynamics supplies is which constant appears, how large it is, and the requirement that the pair be regenerated on the same molecule.

What would settle it. The criterion is addressable in vitro before it is addressable in a neuron. Covalently linked donor–bridge–acceptor radical pairs have tunable exchange coupling and lifetime and are read by transient EPR and optical or CIDNP spectroscopy[2,14,41]; cycling such a system repeatedly and asking whether the product yield of turnover k depends on the input at turnover $k-1$, $k-2$, … measures the memory horizon directly, and varying the interval tests the criterion where it is sharpest. The product's proton T_1 at 50 μT is measurable by relaxation dispersion, and the turnover interval of a candidate flavoenzyme cycle by steady-state kinetics.

Relation to prior work. Quantum reservoir comput-

ing in spin ensembles is established[13,26–29], dissipation is treated as a resource[42], memory beyond the coherence time has been obtained by keeping the record outside the coherent state[43,44], the value of entanglement depends on the input rate[45], and a correlated-spin reservoir with full quantum readout has outperformed large classical ones[46]; our setting differs in that the readout is what chemistry can carry away and the memory sits in a nuclear register. Wong *et al.*[35] asked whether nuclear polarisation accumulates across cryptochrome photocycles and found that an idealised gain does not survive conditions approaching those in vivo; our criterion states what would have to be true for it to survive. Photo-CIDNP has been observed in flavoproteins[40,53,54] though not in cryptochrome. The decoherence objection to quantum-brain proposals[9] concerns ion positions, not nuclear spins; the objection that applies here is nuclear relaxation in the product, and its record is discouraging: the ^{31}P coherence times behind Ref.[6] are estimates since bounded at minutes to seconds[36], and pyrophosphate singlet order relaxes faster than T_1[33]. Thermal modulation of the hyperfine tensors[47,48], decoherence-enhanced sensing[49], Zeno-protected tightly coupled pairs[50], the estimated FAD–TrpH couplings[51] and sixty-spin tensor-network propagation[52] each bound what a fixed-tensor, five-spin model can say, and are the reasons the premise and the tensors are computed in Sec. III F rather than assumed.

Limitations. The model is a five-spin reduction with spin-1/2 nitrogen; the tryptophan nuclei beyond Hβ1 and H1 are not treated, only degree-2 nonlinear capacity is counted, the headline capacities overstate the full register under the published tensors, and the closed cycle assumes that molecules unreacted at τ hand on the nuclear state they then have. Whether a neuronal cryptochrome forms a separated radical pair at all is unestablished, and a behavioural test of magnetic-field effects in *Drosophila* found none[55]; the biological reading is conditional on that, the spin-dynamics results are not.

V. CONCLUSION

Whether a cryptochrome-type radical pair can compute reduces, once the chemistry is left in, to a criterion with three measurable constants. The memory is nuclear and survives recombination, so the clock is the turnover interval and the product's nuclear polarisation reads the memory out, at a capacity of 4.4 against a floor of 2.0 with no spectroscopy. The same proton register must be reused from one turnover to the next, and its relaxation at the geomagnetic field, predicted rather than assumed, caps the memory horizon at 1.7–2.6 ms and at about 1 ms once the cycle's molecule budget is closed, below the neural band unless the register reorients four to fifteen times faster than the protein. The readout is counting-noise limited below about 10^9 molecules per pool, the drive must be catalytic, and no quantum advantage over a matched classical network survives a fair regularisation. The turnover interval, the product's proton T_1 at 50 μT and the pool size are measurable in vitro, before any neuron is involved.

SUPPLEMENTARY MATERIAL

See the supplementary material for S1, liouville-space solver and the coherent limit; S2, capacity estimator and its controls; S3, the injection map; S4, chemically accessible readout routes; S5, coherence fraction at matched hopping rate; S6, classical baselines; S7, turnover clock and the register-reuse criterion; S8, a semiclassical reference for the coherence fraction; S9, which nuclei can hold the register; S10, ensemble averaging; S11, anisotropic hyperfine coupling and the nuclear spin quantum number; S12, predicting the two unmeasured constants; S13, spin parameters; S14, dependence of the cryptochrome point on the electron–electron coupling; S15, reservoir protocol and operating points; S16, use of generative AI tools.

DATA ACCESSIBILITY

All analyses are implemented in the open `qbscreen` package, `https://github.com/deeptell-inc/new_Q_brain`; numerical outputs are in `simulation_results/`, and a module-by-module map is given in the supplementary material. The capacity bound and the solver's agreement with the closed-form coherent limit are checked by the suite in `qbscreen/tests/` (315 tests, all passing). Thirty-seven tests in `test_panel_claims.py` bind the headline quantities and load-bearing claims, and a further one hundred and forty-four in `test_table_rows.py` bind the capacity and standard-deviation cells of every table in the main text and the supplementary material, with the expected values transcribed from the manuscript rather than from the data files; the opposite direction is checked by `test_printed_cells.py`, so the suite fails if either side moves. The LaTeX sources these tests read accompany the submission with the freeze manifest and are not distributed in the public repository; on a clone those tests skip rather than fail.

AUTHORS' CONTRIBUTIONS

H.W. conceived the study, developed the master-equation and reservoir code, ran the simulations, and wrote the manuscript. T.T. contributed to the spin-dynamics formulation and the analysis of the nuclear-register criterion. Both authors reviewed the results and approved the manuscript.

COMPETING INTERESTS

We declare we have no competing interests.

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USE OF GENERATIVE AI TOOLS

Claude Opus 5, Opus 5.2 and Fable 5.1 (Anthropic) and GPT-5.6 (OpenAI) were used to write the `qbscreen` package under H.W.'s direction and review, to draft and revise the text from the authors' outlines and results (every sentence read and approved by the authors), and to review the analysis adversarially before submission, the authors adjudicating every disagreement (supplementary material, Sec. S16). No AI tool is an author; the authors take full responsibility for the content.

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All numerical results are reproducible from the open-source `qbscreen` package.

Appendix A: Model and methods

1. Reservoir Hamiltonian

The reservoir is $N = 5$ spins, two radical-pair electrons (e_1, e_2) and three spin-1/2 nuclei, with Hamiltonian (in MHz)

$$H = \omega_e\,(S_z^{e_1} + S_z^{e_2}) + \sum_{\mu=x,y,z}\Big[A_P\,S_\mu^{e_1} I_\mu^{P} + A_{H_1} S_\mu^{e_1} I_\mu^{H_1} + A_{e_2} S_\mu^{e_2} I_\mu^{H_2} + J\,S_\mu^{e_1} S_\mu^{e_2}\Big], \tag{A1}$$

where $\omega_e = g_e \mu_B B/h$ is the electron Zeeman frequency; each radical carries its own nuclei. At the cryptochrome point the nuclei carry the couplings of ^{14}N5 and ^{14}N10 on FAD$^{\bullet-}$ and Hβ1 on TrpH$^{\bullet+}$ (supplementary material, Table S29), $B = 50$ μT, $\tau = T_2^e = 1$ μs and $J = 0$ as the reference premise; Sec. III F reruns every quantity with the reported couplings[23], D entered as the axial tensor $2D(\boldsymbol{S}_1\cdot\hat{n})(\boldsymbol{S}_2\cdot\hat{n}) - \frac{2D}{3}\boldsymbol{S}_1\cdot\boldsymbol{S}_2$. At the engineered point we use $A_{e_1 a} = 80$, $A_{e_1 b} = 40$, $A_{e_2 a} = 15$ MHz, $J = 2$ MHz and $B = 1$ mT (supplementary material, Sec. S15).

2. Master equation and propagator

Each reservoir update is the completely positive, trace-preserving map $\rho \mapsto e^{\mathcal{L}\tau}\rho$ generated by the Haberkorn master equation[17,30] with electronic pure dephasing,

$$\frac{d\rho}{dt} = -i[H,\rho] - \tfrac{1}{2}\{k_S P_S + k_T P_T, \rho\} + \sum_i \frac{2}{T_2^e}\big(S_z^i \rho S_z^i - \tfrac{1}{4}\rho\big)\,, \tag{A2}$$

solved exactly in Liouville space (H in MHz is multiplied by 2π; rates in μs^{-1}); in the limit $1/T_2^e \to 0$, $k_S = k_T$ it reproduces the closed-form coherent singlet yield to better than 10^{-9} (supplementary material, Sec. S1). Every chemical-readout, clock and criterion result uses $k_S = 1$, $k_T = 0.2$ μs^{-1}.

3. Reservoir protocol

At step k the nuclei are retained and the electron pair is recreated as a spin-correlated pair whose singlet character encodes the input,

$$\rho_e(s_k) = s_k\,|S\rangle\langle S| + (1 - s_k)\,P_T/3, \tag{A3}$$

so that the state entering cycle k is $\rho_e(s_k) \otimes \sigma_{\mathrm{nuc}}^{(k)}$; it evolves for τ under $e^{\mathcal{L}\tau}$, the observables are read out, and a linear ridge readout ($\lambda = 10^{-6}$ unless stated) is trained on them. Inputs s_k are i.i.d. uniform on $[0,1]$; the first 80–100 steps are discarded as washout, so the capacities are independent of the initial spin state. The survivor register hands on the renormalised nuclear state of the pairs that had not recombined by τ; the product register hands on the nuclear state carried away by the recombined molecules,

$$\sigma_{\mathrm{nuc}} \propto \int_0^\tau \big[k_S\,\mathrm{Tr}_e(P_S \rho P_S) + k_T\,\mathrm{Tr}_e(P_T \rho P_T)\big]\,\mathrm{d}t, \tag{A4}$$

with the electron pair reborn unpolarised; the closed carrier sums both branches with their molecule-number weights. The number of input realisations is not uniform and is stated with each result: 12 for the turnover-clock scan and its paired test, which carry the central claim; 8 for the classical baselines and the T_2^e scan at the engineered point; 6 for the headline capacity, the chemical-readout survey, the route floors, the delay kernel, the grid scans, the nuclide scan and the cryptochrome-point classical baselines; 4 for the register-reuse and nuclear-channel scans and the semiclassical reference; and 3 for the ensemble and anisotropy scans. Four quantities are single-realisation: the ridge and shot-noise sensitivity scans and the semiclassical integration-step check, which compare variants on one fixed input by design, and a single-realisation clock scan retained as a control. The single-electron reset of Ref.[26] is used only as a control: for a separated pair it pins the singlet probability at 1/4 and gives zero capacity (supplementary material, Sec. S3). Operating points and convergence checks are in supplementary material Sec. S15.

4. Capacity measures

The linear memory capacity[31] MC $= \sum_{d\geq 0} r^2(s_{k-d})$ sums the squared multiple correlation for reconstructing delayed inputs, each fitted on the first half of a run and scored on the second; the information processing capacity[32] adds degree-2 Legendre targets. The bound IPC $\leq n_{\mathrm{obs}}$ is used as a correctness anchor only; it is asymptotic and exceeded by a few per cent in finite samples (supplementary material, Sec. S2). A route's memoryless floor is the same estimator with the register randomised every cycle ($q = 1$).

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